The Illusion of Calibration:

Vulnerabilities of Unsupervised Test-Time Adaptation in Cross-Ethnic Glaucoma Screening

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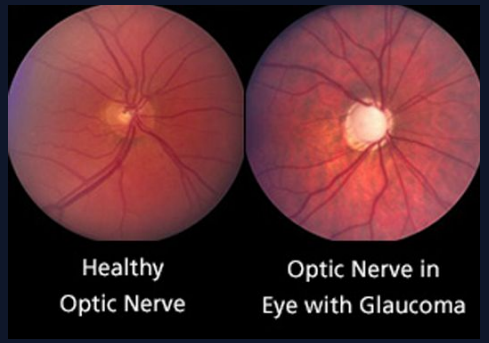
The Core Problem: Cross-Ethnic Domain Shift

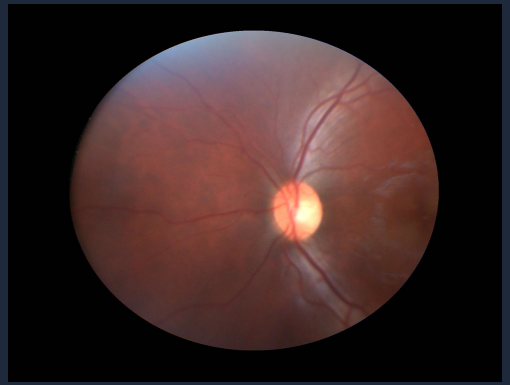
Idea - Do glaucoma screening models trained on Western data (desktop cameras, lighter pigmentation) degrade significantly when deployed on Indian retinas ?

- Shift in acquisition hardware (Desktop vs. Handheld)
- Significant melanin variations in Indian fundus tessellation



Western (AIROGS)

Indian (Remidio Handheld)



Indian (Bosch Handheld)

Our Initial Motivation, Goal & Approach

Primary Goal

Adapt a Western-trained AI to Indian clinical data without needing expensive doctor labels in India.

Idea from - Wang et al., IOVS (2025), “Domain Shift-Induced Unfairness in AI for Diabetic Retinopathy Screening

Tian et al., arXiv (2024), “FairDomain: Achieving Fairness in Cross-Domain Medical Image Segmentation and Classification.”

Technical Approach

Foundation Model - Leveraging DINOv3 (ViT-Large having 300 million parameters) for robust geometric features.

SFDA Paradigm - Source-Free Domain Adaptation via MixEnt-Adapt style injection.

“Prior work shows that retinal AI models suffer significant performance degradation and fairness disparities under cross-dataset and cross-population shifts, often disproportionately affecting underrepresented groups. These findings motivate the need for domain adaptation or retraining when deploying models across populations.”

Initially Proposed Pipeline Journey

1. Pretrain

Source Domain

Train DINOv3 (ViT-L) on Western data (AIROGS) to learn expert-level glaucoma features.

2. Evaluate

Zero-Shot Transfer

Deploy directly on Indian data (Chákṣu). Performance must be degradading due to melanin and hardware shifts.

3. Adapt

Netra-Adapt

Apply MixEnt-Adapt: Unsupervised test-time style calibration using entropy-guided statistics injection.

4. Compare

The Oracle

Compare results against a model fully trained on labeled Indian data.

Dead End 1: The RAG Baseline

The Idea

Can we use the Foundation Model as a search engine? Find similar European images to the Indian input and vote on the diagnosis.

The Result: Failure

AUROC: 0.509

Statistically equivalent to random chance.

The Lesson

Pure visual feature similarity cannot bridge the ethnic gap for subtle pathology.

Attempting Test-Time Adaptation (Our Main Route)

Information Maximization & Distribution Matching

Entropy Minimization

Forces the model to be decisive by making confident guesses on Indian samples.

Diversity Maximization

Prevents mode collapse by ensuring the model utilizes all output classes effectively.

KL-Prior Matching

Strictly enforces Indian disease prevalence (85/15) to align predictions with local demographics.

Dead End 2: The Catastrophic Collapse

The Expectation

Adaptation should push AUROC from 0.852 closer to the Oracle's 0.892.

Hypothesis

Refining the model on unlabelled Indian data should bridge the ethnic feature gap.

The Reality: Collapse

0.390

AUROC: Worse than random guessing!

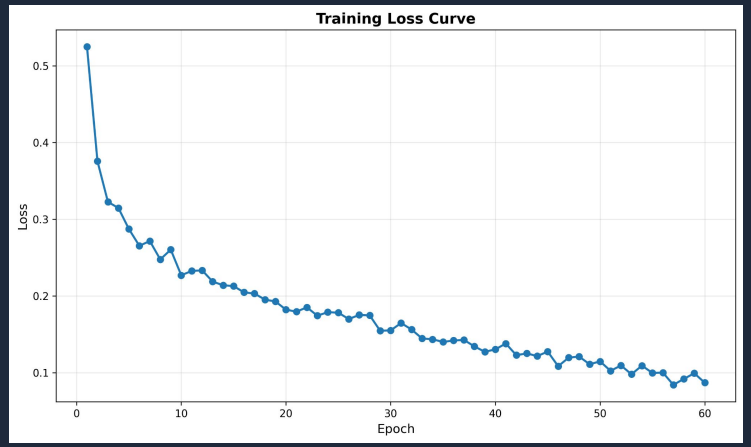
The Failure Point

Specificity dropped to **1.4%**.

The model collapsed into a single-class predictor, failing to distinguish pathology.

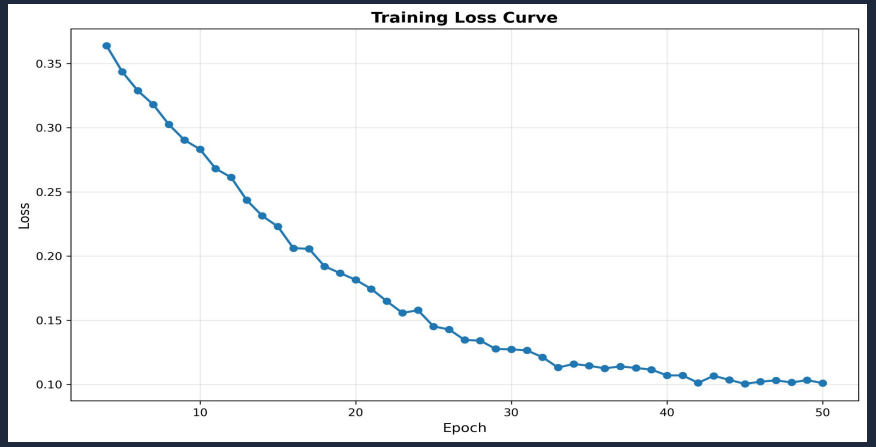
The Mystery: Why did it fail?

Source Training Loss (AIROGS)



The loss curve went down **smoothly**. Mathematically, the model believed it was optimizing perfectly.

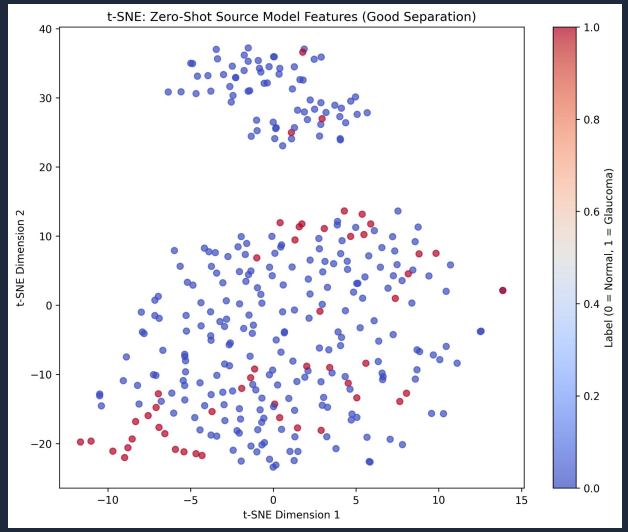
Clinical Performance (The Illusive Reality)



While the loss dropped, it was **destroying the decision boundary**, rendering the model clinically useless. It exploited **confounding variables** (lighting/metadata) rather than glaucoma pathology to reach its mathematical goal is what we think.

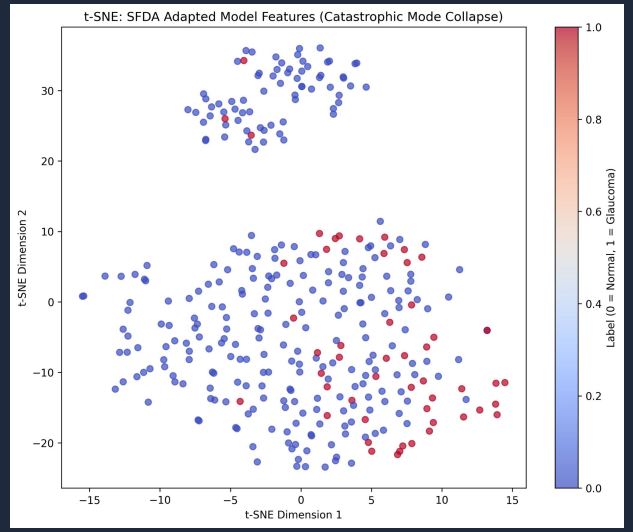
Visualizing the Collapse

Zero-Shot: Structured Features



Clear decision boundaries between healthy and pathological features.

Post-Adaptation: Feature Blur

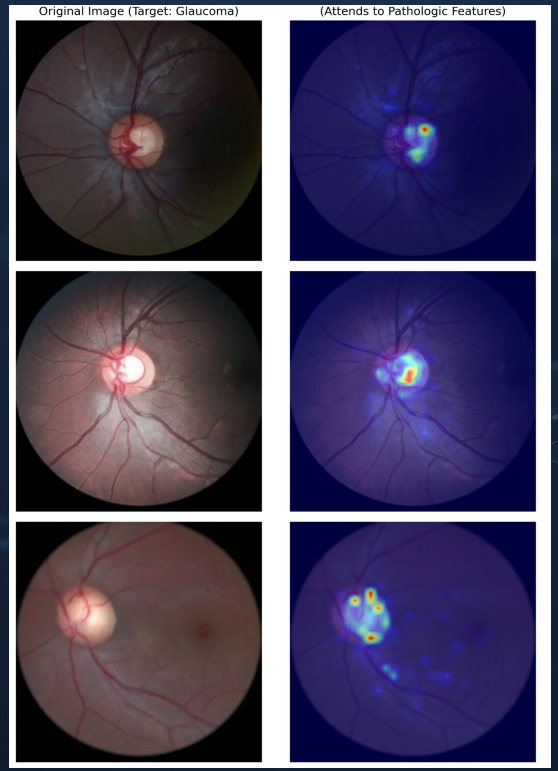


The model collapses into a **single-class predictor**, destroying the clinical boundary.

The Attention Gets Confused

During adaptation, the model stopped looking at the **optic nerve** (the relevant clinical feature) and drifted toward random artifacts.

It began attending to **background noise and style** instead of pathology, explaining the catastrophic performance crash.



The Surprising Good News (Zero-Shot)

Zero-Shot Performance

AUROC: 0.852

Out-of-the-box performance on Indian data without any adaptation.

Compared to Oracle: 0.892

The source model achieves results remarkably close to a model fully trained on Indian labels.

Key Takeaway: Foundation models are incredibly robust learners compared to older CNNs, showing inherent cross-ethnic generalizability.

Conclusion & Clinical Implications

Foundation models generalize incredibly well zero-shot.

Strong pre-training provides robust initial representations of medical pathology before any adaptation occurs.

Unsupervised TTA is currently unsafe for clinical deployment!

Statistical matching of demographics can **destroy decision boundaries** by exploiting confounding variables like style and metadata.